MODEL CARD

AI Model Documentation · FIAR / CIIA UFMG

Model name:

Respiratory Diseases Predictor for SUS

SECTION 0

MODEL SUMMARY

Model Name	Respiratory Diseases Predictor for SUS (RDPSUS)
Version	2.0
Type / Architecture	Linear Regression (baseline) and Gradient Boosting Decision Tree (LightGBM)
Main Goal	Regression
Developer(s)	· Researcher 1, Manager · Researcher 2, Manager · Researcher 3, Contributor
Release Date	27/02/2026
Training Date	26/02/2026
License	FIAR restricted use
Link / Repository	https://github.com/niar-saude-ufmg/SBCAS_26_Respiratory_Disease/v2.ipynb

SECTION 1

INTENDED USE AND CONTEXT

1.1 Description and Motivation — OBLIGATORY

Model Description	Forecasting model, predicting the total hospitalizations by respiratory diseases for a given hospital and month. For comparison purposes, 4 models were implemented: a baseline (Linear Regression, with "scikit-learn"), LightGBM (model by Microsoft), LightGBM with leakage, and LightGBM after resampling.
Motivation	Research for testing the viability of the FIAR workflow on an experimental model.

1.2 Users and Use cases — OBLIGATORY

Expected Users	FIAR Researchers

Groups Impacted by the Model	Groups under subcategories of age, sex and race; and hospitals in different regions.
Intended Use Cases	Research for FIAR researchers only.
Unintended Use Cases	Clinical decision making without medical supervision; use in populations outside training scope; use as a medical device without registration in ANVISA; Use for assistance on individual cases (patients).
Implementation Conditions	Research

⚠ **AI Models applied to direct clinical decisions can be classified as medical devices by ANVISA (RDC 657/2022). Verify the need for a registration before implementing the model in a care environment.**

SECTION 2

TRAINING AND EVALUATION DATA

2.1 Training Data

OBLIGATORY

Used Dataset(s)	SIHSUS - Hospitalizations for Respiratory Diseases; version 2.0.
Data size	103.846 instances. The dataset was split for training (2022 - 06/2024), validating (07/2024 - 12/2024) and testing (01/2025 - 11/2025).
Time Window	01/2022 - 11/2025
Pre-processing	Removed instances with null values and split the dataset for training, validating and testing. For fairness evaluation, SMOTENC was used for balancing instances based on regions.
Data Privacy and Protection	Data is aggregated by hospital, with no individual cases present in the data. Both the dataset and models can only be accessed by FIAR researchers.
Training Data Limitations	People of age between 15 to 59 and people of color are underrepresented, as well as hospitals from the North region. A bias evaluation was done and can be found on the Fairness Report.

2.2 Evaluation Data

OBLIGATORY

Used Dataset(s)	SIHSUS - Hospitalizations for Respiratory Diseases; version 2.0. The dataset was split for training, validating and testing.
Data Size	36.786 instances.
Representativeness	The test dataset represents the same groups covered by the training split.

2.3 Validation Data

OBLIGATORY

Used Dataset(s)	SIHSUS - Hospitalizations for Respiratory Diseases; version 2.0. The dataset was split for training, validating and testing.

Data Size	20.952 instances.

SECTION 3

TRAINING AND ARCHITECTURE

3.1 Technical Specifications

OBLIGATORY

Framework / Libraries	pandas 3.0.1; numpy 2.4.2; matplotlib 3.10.8; scikit-learn 1.8.0; geopandas 1.1.2; jupyter 1.1.1; lightgbm 4.6.0; shap 0.50.0; imblearn 0.0.
Main Hyperparameters	For LightGBM, objective: poisson; seed: fixed random seed; early stopping in 5 rounds; and validating dataset. Due to LightGBM's reliance on a random value, it was trained 5 times and averaged out, each with a different random seed (778, 768, 758, 748, 738).
Training Hardware	12th Gen Intel® Core™ i5-1235U × 12 cores CPU.
Training Environment	On-premise training, closed environment.
Training Time	20 minutes on average.
Validation strategy	Dataset split for LightGBM validation with early stopping.

3.2 Influencing Factors in Performance

OBLIGATORY

Group Factors	Performance may vary depending on state and hospital size
Instrumental Factors	Performance may vary depending on quality of medical records
Environment Factors	Performance may vary depending on disease seasonality, pandemic events, changes in clinic protocols or ICD-10 vs. ICD-11

SECTION 4

PERFORMANCE EVALUATION

4.1 Used Metrics

OBLIGATORY

Primary Metrics	SMAPE (Metric for evaluating the average error, percentage wise).
Secondary Metrics	MAE (Basic metric for evaluating average error), RMSE (Metric for evaluating average error, while penalizing bigger errors).

4.2 General Results

OBLIGATORY

Metric	Model	Result
SMAPE	Baseline (Linear Regression.)	43.2907
SMAPE	LightGBM (Base)	32.2883

SMAPE	LightGBM with Fairness	26.2784
SMAPE	LightGBM with Leakage	31.9723
RMSE	Baseline (Lin. Regres.)	14.1346
RMSE	LightGBM (Base)	13.8134
RMSE	LightGBM with Fairness	12.4684
RMSE	LightGBM with Leakage	14.1899
MAE	Baseline (Lin. Regres.)	8.71728
MAE	LightGBM (Base)	8.34667
MAE	LightGBM with Fairness	7.14794
MAE	LightGBM with Leakage	8.31573

4.3 Performance by Group

OBLIGATORY

⚠ **CRITICAL FIELD for Responsible AI — evaluate the model separately for relevant populations.**
Priority groups in health care: age group · sex · race/color · region (city, states) · type of coverage (SUS / health insurance) · hospital size.

Group	Category	Primary Metric	Result
0-14	Age	SMAPE	30
15-59	Age	SMAPE	34.8
60+	Age	SMAPE	32.6
Male	Sex	SMAPE	31.2
Female	Sex	SMAPE	33
White	Race	SMAPE	30.5
Black	Race	SMAPE	36.4
Others	Race	SMAPE	32.9
AC	State (UF)	SMAPE	33.7
AL	State (UF)	SMAPE	31.7
AM	State (UF)	SMAPE	33.1
AP	State (UF)	SMAPE	36.7
BA	State (UF)	SMAPE	36
CE	State (UF)	SMAPE	32.4
DF	State (UF)	SMAPE	26.7
ES	State (UF)	SMAPE	30.6
GO	State (UF)	SMAPE	35.2
MA	State (UF)	SMAPE	37
MG	State (UF)	SMAPE	30.5
MS	State (UF)	SMAPE	36.8

MT	State (UF)	SMAPE	31.5
PA	State (UF)	SMAPE	33.7
PB	State (UF)	SMAPE	30.2
PE	State (UF)	SMAPE	33
PI	State (UF)	SMAPE	32.3
PR	State (UF)	SMAPE	32.7
RJ	State (UF)	SMAPE	30.5
RN	State (UF)	SMAPE	38.8
RO	State (UF)	SMAPE	35.4
RR	State (UF)	SMAPE	43.9
RS	State (UF)	SMAPE	32.7
SC	State (UF)	SMAPE	31
SE	State (UF)	SMAPE	31
SP	State (UF)	SMAPE	25.6
TO	State (UF)	SMAPE	36.7
Small	Hospital Size	SMAPE	43.7
Medium	Hospital Size	SMAPE	35
Large	Hospital Size	SMAPE	28.8
Extra Large	Hospital Size	SMAPE	24.3

4.4 Error analysis — OBLIGATORY

Observed Error Patterns	Disparities were found inside Regional groups and Hospital sizes. In the former, states from the north region got a higher error average, while SP and DF obtained the best results. In the latter, a correlation can be seen, where the smaller the hospital is, the harder the model found it to be predictable.

SECTION 5
EQUITY AND ETHICAL CONSIDERATIONS

5.1 Ethical Impact — OBLIGATORY

Expected Positive Impacts	Hospital management decisions.
Expected Negative Impacts	Automation of decisions that should have human supervision, reduction of patient autonomy, missed diagnosis can lead to poor allocation of resources.
Populations in risk	This model affects everyone and presents a risk for everyone, but more specifically for regions with smaller hospitals.

5.2 Bias and Equity Evaluation — OBLIGATORY

Evaluated Sensitive Features	Race/Color, Sex, Age, Region and Hospital Size.
Used Equity Metrics	SMAPE was used for evaluation by comparing model errors for each individual group. For more details, read the Fairness Report.
Equity Evaluation Results	The model showed demographic parity between sex , race and age groups, but there was a gap of 18.8% in sensitivity for smaller hospitals and states from the north.
Adopted Mitigations	Using SMOTENC, a class balancing for all regions was done. For more details, read the Fairness Report.
Known Residual Biases	Although reduced, all previous bias still persisted (10.29%).

5.3 Regulatory Compliance

OBLIGATORY

Risk Classification *CFM N° 2454*	Low. The model is used exclusively for research purposes by FIAR researchers (Section 1.2) and is not applied to individual clinical decisions. Its outputs support hospital-level planning, not patient-level intervention.
Ethical Report / CAAE	Not applicable. The research team did not have access to non-anonymized or individually identifiable data at any stage; the SIH-SUS data used was already aggregated by hospital (Section 2.1), which does not constitute human-subjects research requiring CEP/CONEP submission.
Data Protection Impact Assessment (DPIA)	A formal DPIA (LGPD Art. 38) was not conducted. Since only aggregated, non-identifiable hospital-level data was accessed at any point in the pipeline, the processing does not involve personal data as defined by LGPD Art. 5, reducing the associated risk that would trigger a mandatory DPIA.
Fundamental Rights Impact Assessment (FRIA) / (AIA)	Not conducted in this version. The model is currently restricted to a research context (Section 1.2) and is not used in direct clinical decision-making. An AIA (CFM Nº 2.454, Anexo I, Inciso XIII) would be required before any transition to clinical or operational deployment.
EU AI Act Compliance	Not applicable. The model is developed and operated exclusively within the Brazilian public health system (SUS/DataSUS) and is not placed on the market or put into service within the European Union.

SECTION 6

EXPLAINABILITY AND INTERPRETABILITY

Type of Explainability	Global and Local
Methods and Techniques	For explanation, SHAP was implemented.
User Presentation	An Explainability Report presents the most important features for model prediction. Also, SHAP can be used to indicate, for a data point, which variables most mattered for the model.
Audit Link	https://github.com/niar-saude-ufmg/SBCAS_26_Respiratory_Disease/blob/main/docs/metrics_report.md

SEÇÃO 7

DEPLOYMENT AND OPERATION

7.1 Technical Requirements

OBLIGATORY

Execution Environment	Not applicable. The model has not been deployed to a production or clinical environment; it currently runs only in a closed research environment (see Section 3.1).
Dependencies	See Section 3.1 (Framework/Libraries) for the full list of libraries and versions used during development.
Minimum Hardware Requirements	Not applicable. No dedicated production hardware has been defined, as the model is not deployed outside the research setting.
Inference Time	Not applicable. No inference time or latency constraints have been established, since the model does not operate in a real-time or production pipeline.
Expected Integration	Not applicable. No integration with clinical workflows, hospital information systems, or dashboards has been implemented at this stage.

7.2 Monitoramento e Manutenção

OBLIGATORY

Performance Monitoring	Not applicable. No production monitoring pipeline exists, as the model remains in the research phase (Section 1.2).
Data Drift Detection	Not applicable at this stage. A drift detection methodology would need to be defined prior to any operational deployment.
Retraining Criteria	Not applicable. No retraining schedule or triggering criteria have been defined outside the research/experimentation cycle.
Technical Responsible	Not applicable. No operational technical responsibility has been assigned, as the model has not transitioned to a production or clinical context.

SECTION 8

LIMITATIONS, WARNINGS AND RECOMMENDATIONS

Known Abilities	The model is capable of predicting expected values for next months based on hospital history. If the hospital presents a complete history (preferably, over than a year), then the model is at its ideal condition.
Known Limitations	The model was not tested for hospitals outside of DataSUS, nor populations other than Brazilians (who may present different general characteristics).
Critical Warnings	Performance unstable for small hospitals.
Use Recommendations	Always have it under clinical supervision; review predictions with lower reliability; don’t use it as the single criteria for decisions.

SECTION 9

VERSIONING AND CARD HISTORY

Version	1.1.0
Update Description	Change of Template
Last Modified	20/07/2026
Change History	· v1.0.0 (01/03/2026): Initial release. · v1.1.0 (20/07/2026): Template change.

Adapted from Mitchell et al. (2018) "Model Cards for Model Reporting" · ACM FAccT 2019 · Incorporates practices from Hugging Face Model Card Guidebook, NVIDIA Model Card++, EU AI Act (2024) and CFM Nº 2454.